Data Science Lab Mini Project · Semester VII · 2026-27

KAVACH AI

Voice-Enabled Scam & Fraud Message Detector with Explainable ML

Presented by

Nadar Yuvrajvarthan Murugesh – 123A3032

Akash Sivan Konar – 123A3025

Under the guidance of

Ms. Savita Lohiya

Group 17 · Department of Information Technology · SIES Graduate School of Technology, Nerul

Index

1 Introduction

2 Problem Definition

3 Dataset and Honest Evaluation

4 Hardware and Software Requirements

5 System Architecture

6 Methodology

7 Results

8 Explainability and Shortcut Fixes

9 Application Screenshots

10 Applications

11 Business Intelligence Decision

12 References

13 Conclusion

Introduction

Fraud messages arrive every day

Fake bank KYC alerts, "digital arrest" calls from people posing as police or the CBI, job and investment offers that ask for a fee, parcel and customs scams.

Built to cause panic

Urgency, threats (account blocked, connection cut, arrest), and requests for an OTP, PIN or payment.

Hard to tell apart

Many people, especially elderly and first-time smartphone users, cannot easily separate them from genuine bank or delivery notifications.

Kavach AI

"Kavach" means armour or shield.

- Paste the text
- Upload a screenshot
- Speak or upload a voice note

→ scam type, risk level, the words that gave it away, and what to do next

Lost money to fraud? Call the national cyber-crime helpline **1930** or report at **cybercrime.gov.in**

Problem Definition

1. Input

SMS / WhatsApp text, a screenshot or a voice note

2. Classify

one of 5 classes with a calibrated probability

3. Score

scam probability = 1 − P(legit) and a risk tier

4. Explain & advise

trigger words, red flags and safe next steps

The 5 classes (Sentinel dataset, 14,000 messages)

5,600	2,905	3,080	1,875	540
Legit	Government impersonation	Financial fraud	Bank phishing	Romance / parcel / sextortion

Risk tiers from the scam probability

High	Medium	Low
≥ 0.80	0.50 – 0.80	< 0.50

Classes are imbalanced (smallest = 3.9%), so models are compared on macro-F1, not accuracy.

Dataset and Honest Evaluation

14,000

messages in Sentinel, 5 classes, 0 exact duplicates

6,693

distinct templates: 52% of rows repeat one

3.9%

smallest class (romance / parcel / sextortion): imbalance

2

out-of-domain test sets the model never saw

UCI SMS Spam Collection

5,171 real SMS after removing 403 duplicates. UK spam and casual chat: a tough real-world test.

Hinglish & English scam set

3,558 messages after removing 229 duplicates, but 97.6% are UCI rows, so the 87 Hinglish-only rows are reported separately.



Template-grouped split: digits, links, phones and amounts are masked to a template key, and StratifiedGroupKFold keeps every template on one side, so the test set is never a near-copy of training data. Best model chosen on this split.

Hardware and Software Requirements

Hardware

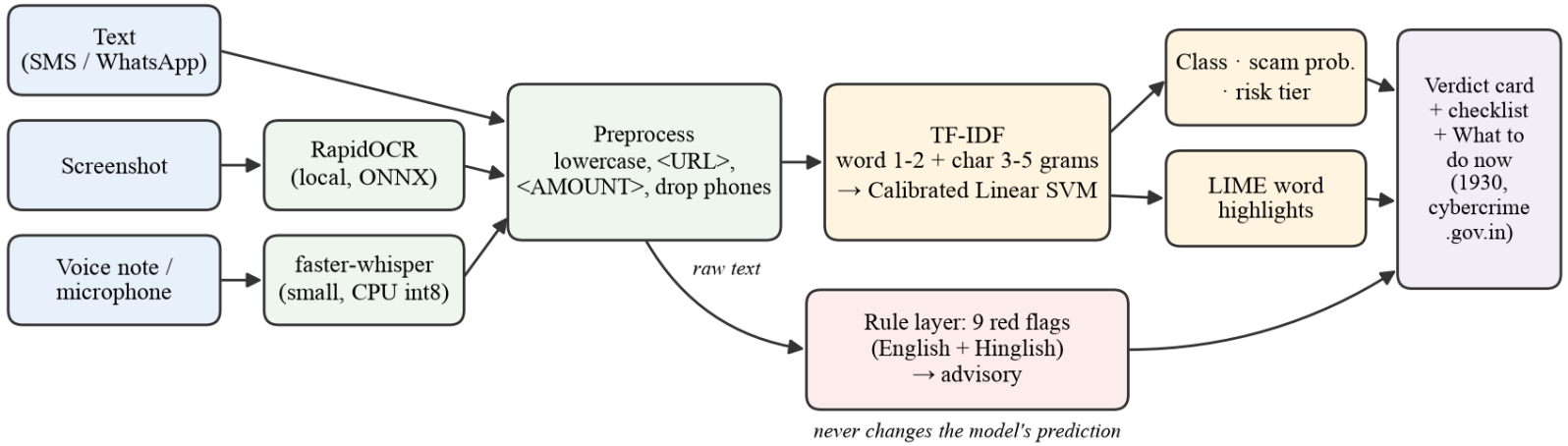
- Tested on a MacBook (Apple M5, 16 GB RAM)
- About 6 GB of disk for the local models (qwen2.5:7b 4.7 GB, Whisper small 0.5 GB)
- Microphone and speakers for voice input and spoken replies
- No GPU, no cloud and no internet needed

Software

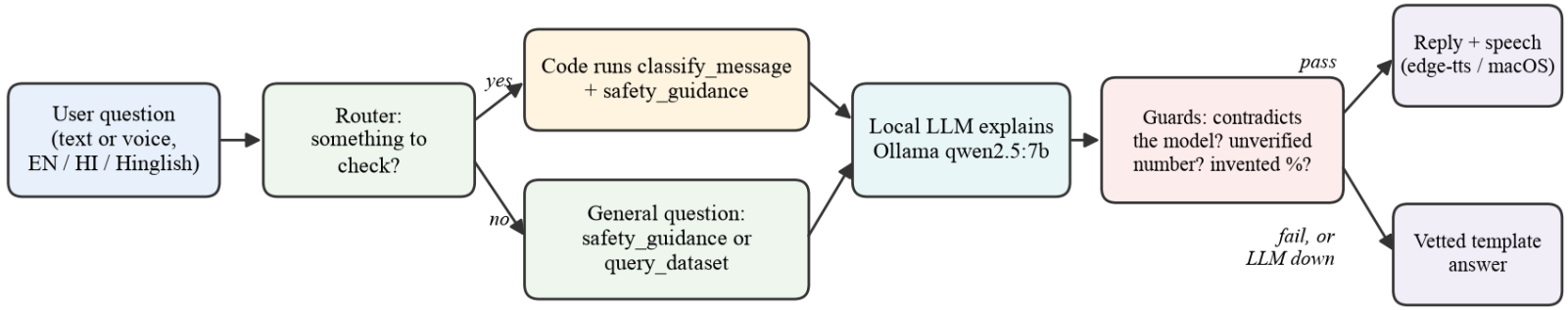
- Python 3.11, pandas, NumPy, scikit-learn, LIME
- Streamlit (multipage web app)
- RapidOCR (ONNX Runtime) for screenshots
- faster-whisper (small, CPU int8) for voice
- Ollama with qwen2.5:7b for the assistant
- edge-tts, with macOS voices offline
- matplotlib, seaborn, Jupyter, pytest

System Architecture

(a) Scan a Message: the trained model classifies



(b) Kavach Assistant: the LLM only explains



Everything runs locally on the laptop: no message leaves the device and nothing is stored.

Methodology

1

Clean the text

Lowercase; links → <URL>, amounts → <AMOUNT>; phone numbers removed; Hinglish words kept.

2

Two test splits

Stratified random 80/20 (seed 42) and template-grouped split: 11,200 train / 2,800 test.

3

TF-IDF features

Word 1-2 grams for phrases + character 3-5 grams for Hinglish spelling variants.

4

Compare 4 models

Naive Bayes, Logistic Regression, Linear SVM (sigmoid-calibrated), Random Forest (200 trees); balanced weights.

5

Explain

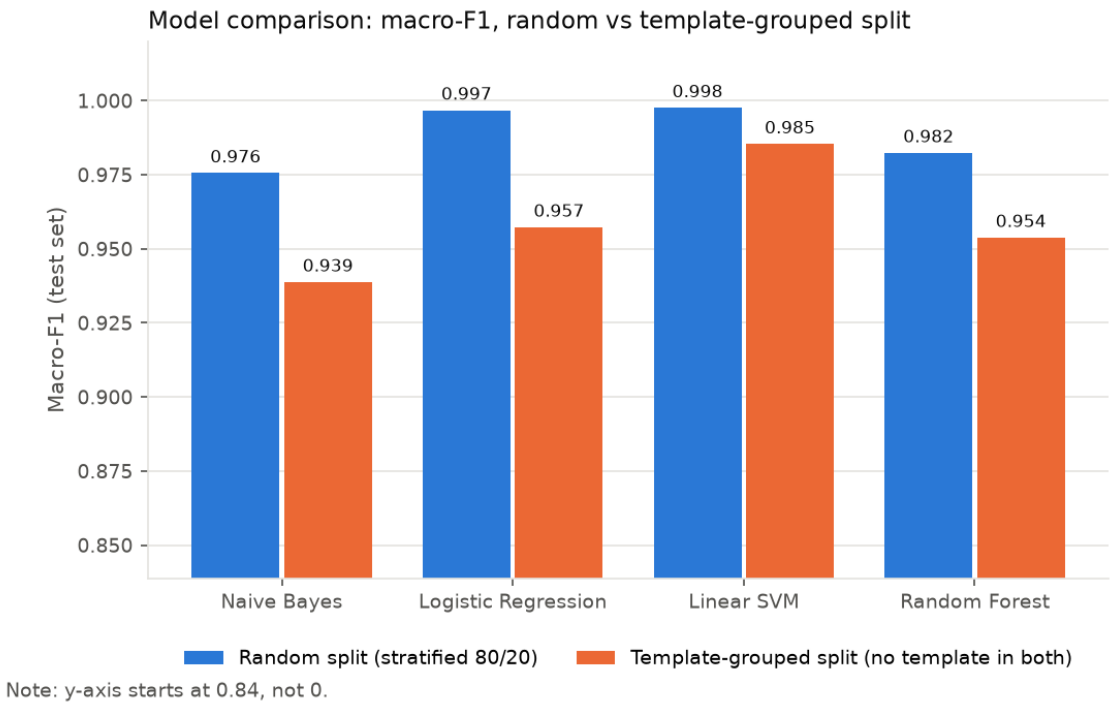
LIME (500 samples, ≈0.03 s) for each message; SVM coefficients for global top words.

6

Rule layer

9 red-flag tactics in English + Hinglish; 3+ = high, 2 = caution; never changes the prediction.

Results: Model Comparison



Note: y-axis starts at 0.84, not 0.

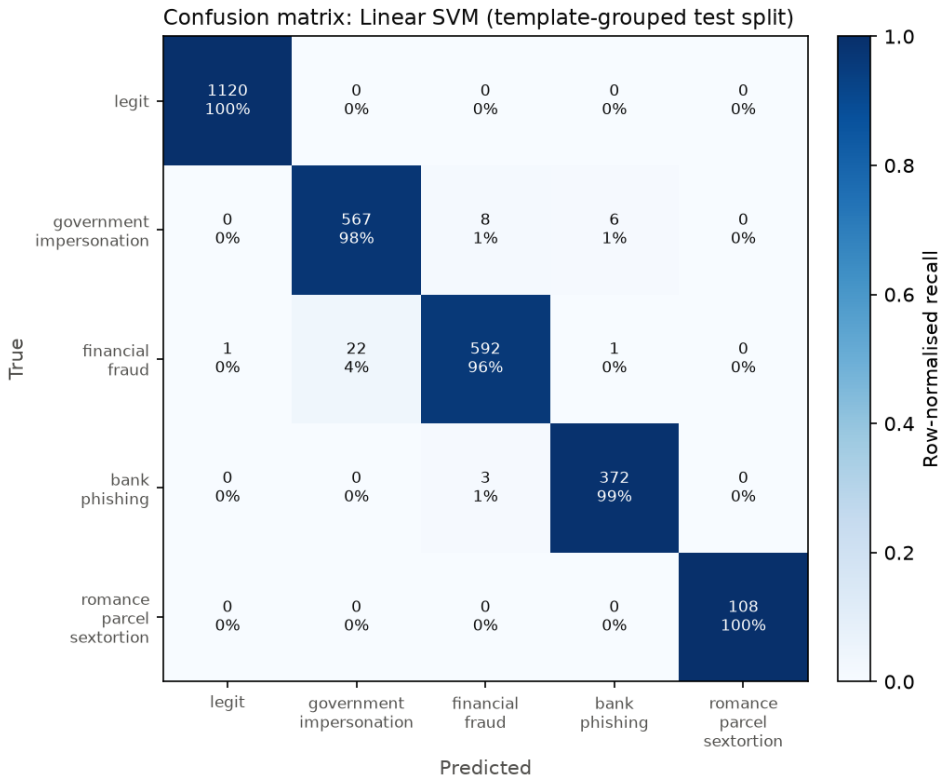
Macro-F1, random vs template-grouped split (y-axis starts at 0.84)

0.9854

Linear SVM macro-F1 on the template-grouped split: our headline number

- Best on both splits; trains in under 3 s
- Random split 0.9976 is inflated by repeated templates
- Next best: Logistic Regression 0.9573

Results: Per-Class Performance

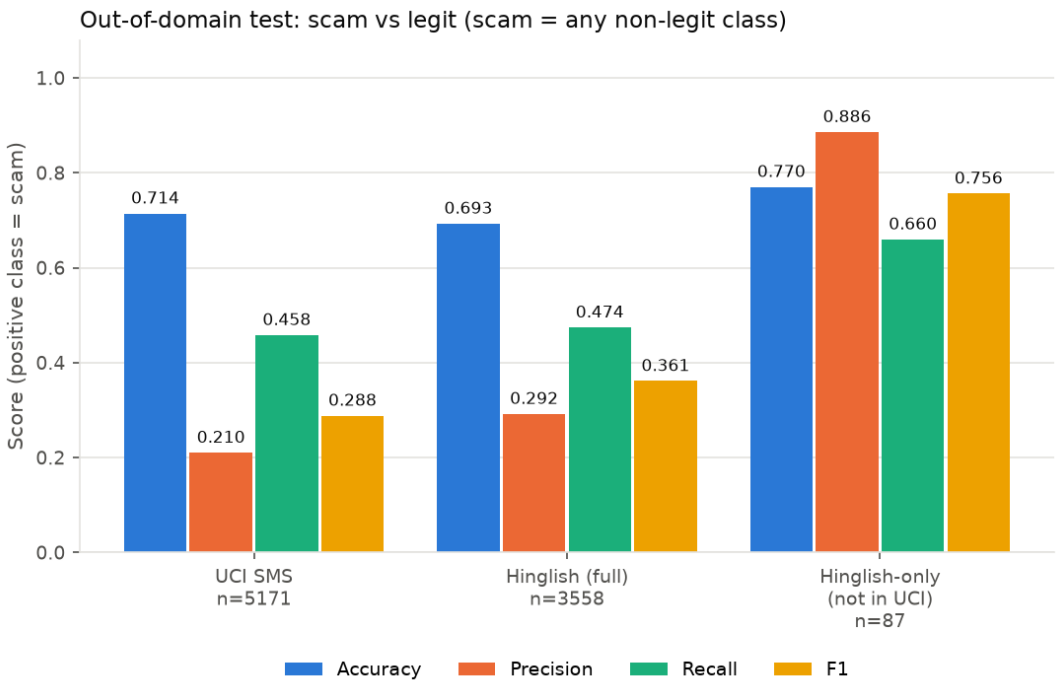


Linear SVM, template-grouped test split

Class	Precision	Recall	F1
Legit	0.9991	1.0000	0.9996
Government imp.	0.9626	0.9759	0.9692
Financial fraud	0.9818	0.9610	0.9713
Bank phishing	0.9815	0.9920	0.9867
Romance / parcel	1.0000	1.0000	1.0000

- All 1,120 legit test messages correct
- Smallest class still reaches F1 1.00 (balanced weights)
- Main confusion: financial fraud ↔ government (22), matching label noise in the data

Results: Real-World (Out-of-Domain) Test



Scam vs legit on data the model never saw (positive class = scam)

Why the scores drop

No casual chat

Sentinel's legit class is only bank, recharge and delivery alerts, so personal SMS gets over-flagged (UCI precision 0.21).

Different scams

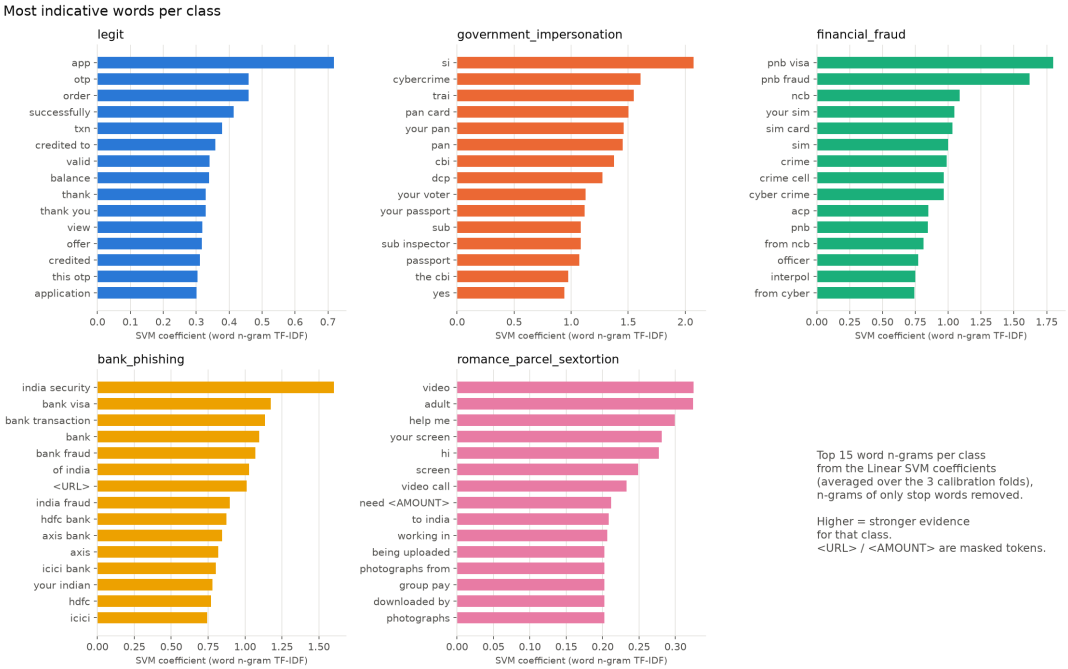
UCI's spam is UK prize and lottery spam, unlike Indian fraud (recall 0.46).

Name shortcuts

The model partly learned bank and agency names, not only tactics.

No tuning was done on these sets, so they stay a fair test.

Explainability and Shortcut Fixes



Most indicative words per class (Linear SVM coefficients)

LIME: for each message, 500 perturbed copies show which words moved the prediction. Red pushes toward scam, green toward legit.

Watch-out: financial-fraud and bank-phishing words are led by bank names ("pnb visa", "hdfc bank").

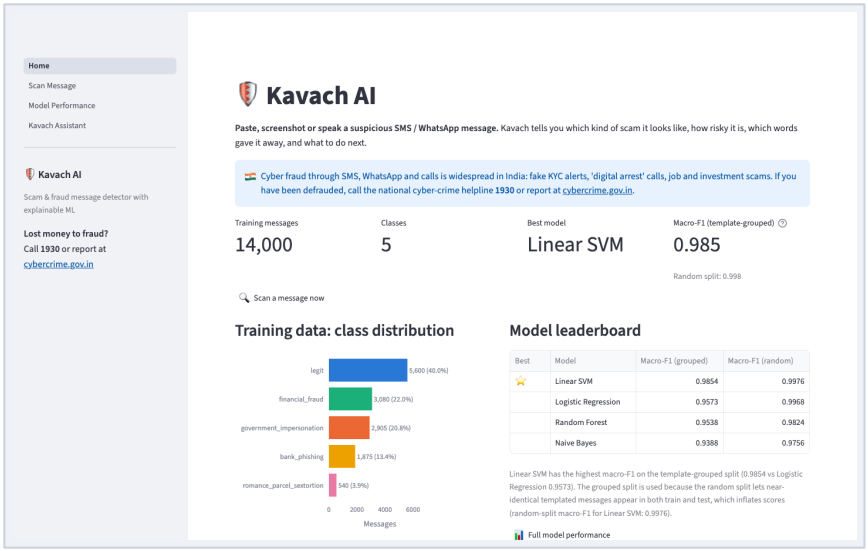
0.38 → 0.46

UCI recall after removing phone numbers (16.2% of legit vs 0% of scam rows had one). In-domain score unchanged.

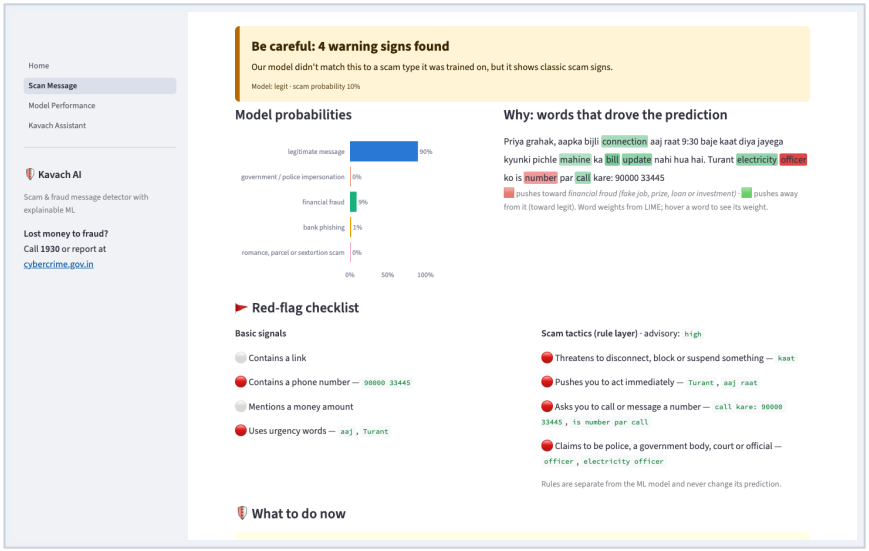
0%

of legit messages get a "high" rule warning (any warning: 6.2% Sentinel, 0.04% UCI).

Screenshots: Home and Scan

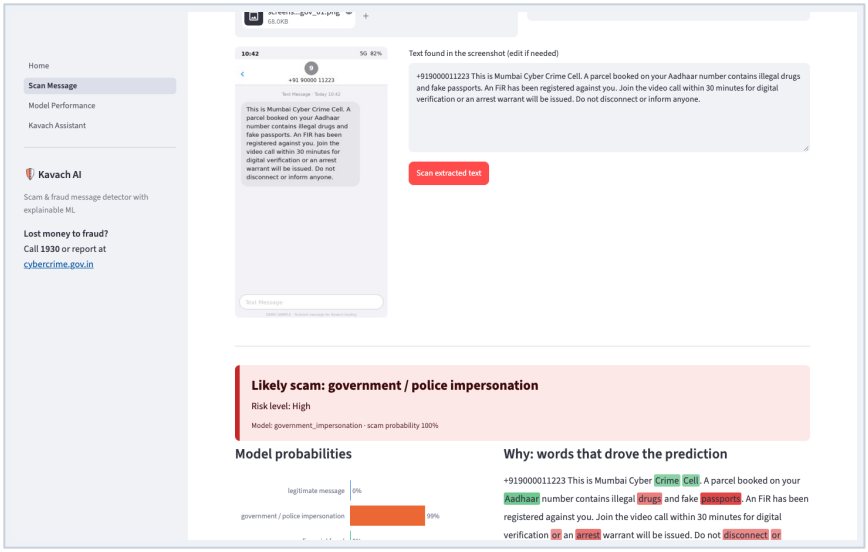


Home: 14,000 messages, 5 classes, Linear SVM, grouped macro-F1 0.985

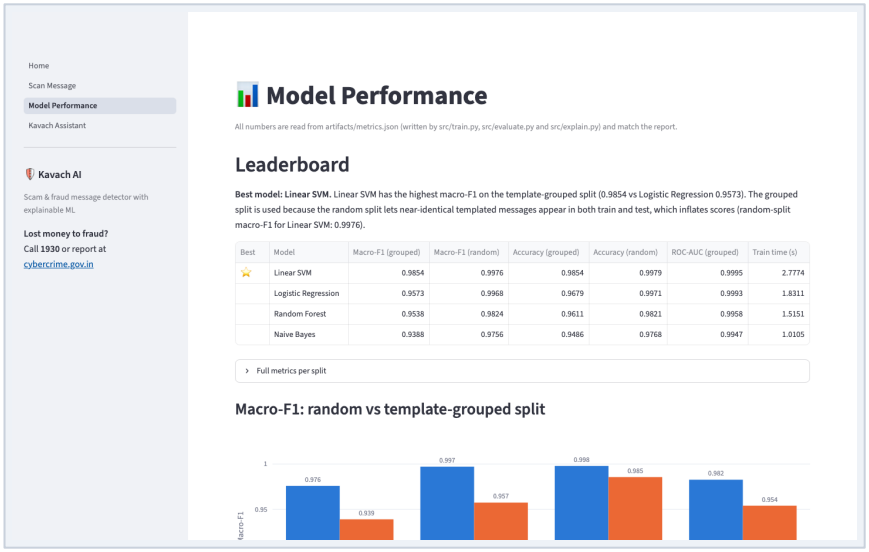


Hinglish electricity scam: model says legit (10%), rule layer finds 4 warning signs → amber card

Screenshots: OCR and Model Performance

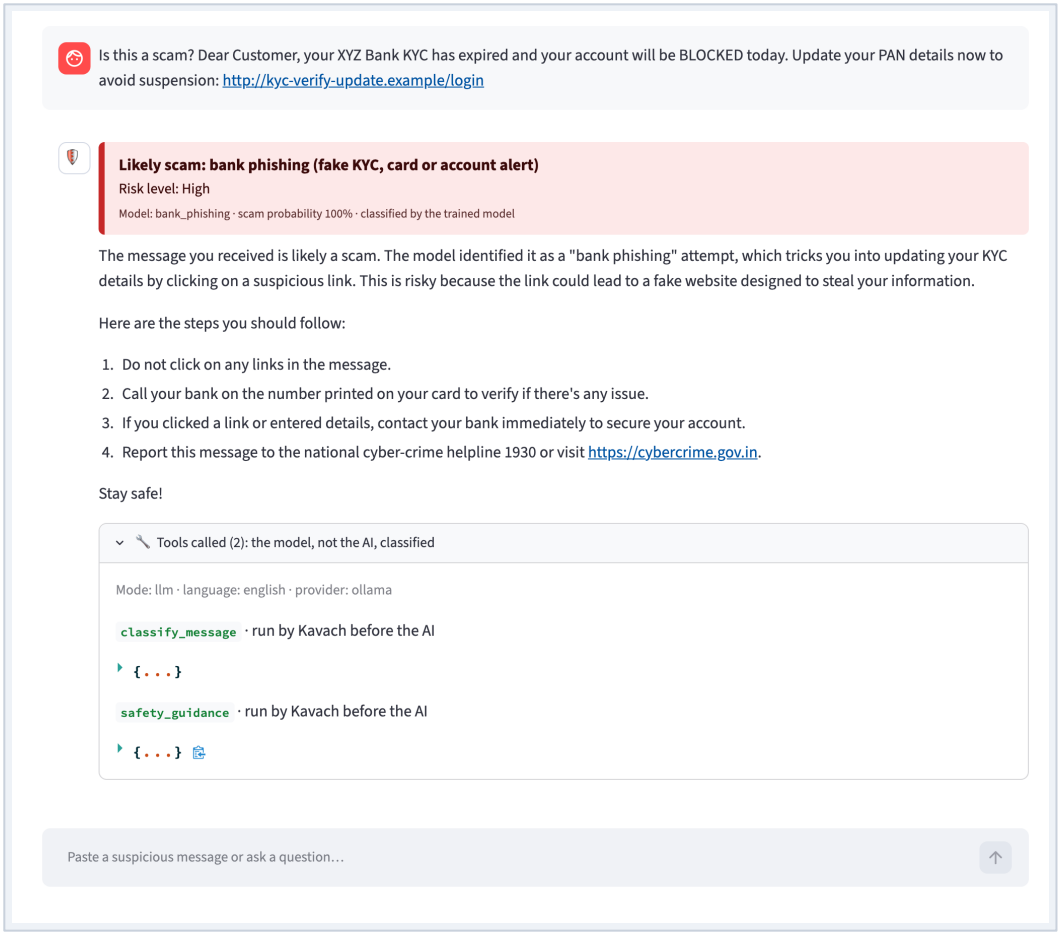


Screenshot of a "digital arrest" message read by RapidOCR → government impersonation, High



Model Performance page: both splits, charts, OOD results and limitations

Screenshots: Kavach Assistant

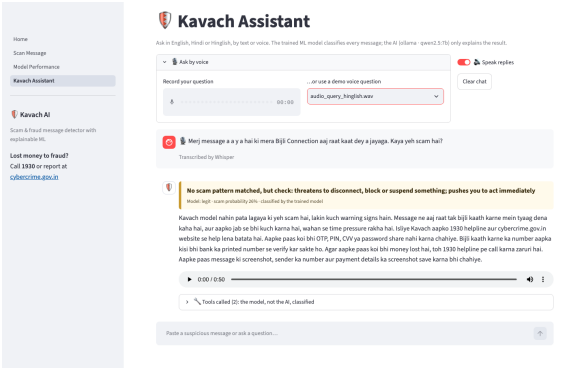


The model classifies, the AI explains

Our code runs classify_message and safety_guidance before the local LLM (qwen2.5:7b via Ollama) writes anything; the tools panel proves it.

Guards

A reply is replaced by a vetted answer if it calls a flagged message safe, invents a number or quotes an unsupported %.



Hinglish voice question, spoken reply

Applications

Personal scam checker

Paste, screenshot or speak a message and get a verdict and safe steps in seconds.

Bank & telecom support

Agents check a forwarded message and explain the red flags to the customer.

Cyber-cell awareness

Explanations and per-class statistics as material for awareness drives.

Digital literacy

Every verdict shows why, so people learn to spot the tactics themselves.

Privacy-sensitive use

Runs fully on the device: messages never leave the machine.

Reusable method

Grouped split, OOD tests and shortcut ablation carry over to other fraud text.

Business Intelligence Decision

High

scam probability ≥ 0.80

Treat as a scam: don't click, call, reply or pay. Block and report on 1930 / cybercrime.gov.in.

Medium

scam probability 0.50 – 0.80

Don't act until verified through an official channel, such as the number on your bank card.

Low

scam probability < 0.50

"No scam pattern detected", never "safe". If the rule layer finds 3+ signs, the card turns amber.

What the data tells awareness campaigns

24.6%

of scams contain a link vs 3.4% of legit

75.6%

of bank-phishing messages contain a link

91.0%

of financial-fraud messages name an amount

80.0%

of government-impersonation messages use urgency words

References (IEEE format)

[1] T. A. Almeida, J. M. Gómez Hidalgo, A. Yamakami, 'Contributions to the Study of SMS Spam Filtering: New Collection and Results,' ACM DocEng, 2011.

[2] S. J. Delany, M. Buckley, D. Greene, 'SMS Spam Filtering: Methods and Data,' Expert Systems with Applications, 39(10), 2012.

[3] T. Joachims, 'Text Categorization with Support Vector Machines,' ECML, 1998.

[4] J. Platt, 'Probabilistic Outputs for Support Vector Machines,' Advances in Large Margin Classifiers, MIT Press, 1999.

[5] M. T. Ribeiro, S. Singh, C. Guestrin, ''Why Should I Trust You?': Explaining the Predictions of Any Classifier,' ACM KDD, 2016.

[6] S. Kaufman et al., 'Leakage in Data Mining: Formulation, Detection, and Avoidance,' ACM TKDD, 6(4), 2012.

[7] A. Radford et al., 'Robust Speech Recognition via Large-Scale Weak Supervision,' ICML, 2023.

[8] K. Greshake et al., 'Not What You've Signed Up For: Indirect Prompt Injection,' ACM AISec, 2023.

[9] Shade63, 'Scam Classification (Multiclass) - Sentinel Dataset,' Hugging Face, 2026.

[10] Qwen Team, 'Qwen2.5 Technical Report,' arXiv:2412.15115, 2024.

[11] F. Pedregosa et al., 'Scikit-learn: Machine Learning in Python,' JMLR, 12, 2011.

[12] Ministry of Home Affairs, 'National Cyber Crime Reporting Portal,' cybercrime.gov.in, 2026.

Conclusion

Trained model decides, every verdict explained

Calibrated Linear SVM, macro-F1 0.9854 (grouped), with LIME words, red flags and vetted advice.

Evaluated honestly

Grouped split instead of the inflated 0.998; OOD gaps measured and explained; a phone-number shortcut fixed.

Practical and private

Text, screenshot or voice in English, Hindi and Hinglish, running entirely on a laptop.

Future work

Fine-tune multilingual DistilBERT · a real Hinglish dataset with casual chat · more scam types (utility, "new number") · relabel noisy classes · user-feedback loop · a mobile / message-forwarding version



Thank You

Questions?

Nadar Yuvrajvarthan Murugesh · Akash Sivan Konar · Guide: Ms. Savita Lohiya

Lost money to fraud? Call 1930 or report at cybercrime.gov.in